

Development of an Instructional Framework for Source Code Documentation in Undergraduate IT Programming Courses based on Diataxis Framework

STATEMENT OF THE PROBLEM

1. What are the challenges faced by IT faculty in teaching source code documentation in terms of:

- 1.1 Instructional delivery
- 1.2 Student compliance
- 1.3 Assessment of documentation

2. What are the current teaching strategies applied by IT faculty to students regarding their source code documentation based on Diátaxis Framework categories in terms of:

- 2.1 Tutorials
- 2.2 How-to-Guides
- 2.3 Explanations
- 2.4 Reference materials

3. What is the level of implementation of current source code documentation practices observed by IT faculty among their students in terms of:

- 3.1 Code commenting
- 3.2 Code readability
- 3.3 Code consistency

4. To what extent do IT faculty apply their industry practices in teaching source code documentation to students in terms of:

- 4.1 Coding standards
- 4.2 Documentation tools
- 4.3 Documentation requirements

5. What instructional framework for source code documentation in undergraduate IT programming courses can be developed based on the findings of the study?

Diataxis Framework

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It is a structured instructional framework focused on organizing software documentation into four distinct categories. These categories are the form of documentation practices to better understand the needs of documentation users and the documentation itself.

Categories:

- **Tutorials** - it is teaching the students through practical hands-on experience. *“The user will learn through what they do - not because someone has tried to teach them.”. It focuses on learning.*
- **How-to-Guides** - it is a goal-oriented instruction for solving specific problems. It tells the student what to do to achieve the expected result. **It focuses on reaching the goal.**
- **Explanation** - it is a conceptual understanding, provides context and background to the topic at hand. **It is about understanding.**
- **Reference** - *“Reference guide offers technical description”*. It **provides information** designed for a quick lookup.

References:

[Diátaxis](#)

[Diataxis Framework Explained by Katara](#)

[diataxis-documentation-framework/how-to-use-diataxis.rst at main ·](#)

[evildmp/diataxis-documentation-framework](#)

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